

EXPERIMENT 05

Use Autopsy to Create a Case and Import Evidence

OBJECTIVE

To create a digital forensics case in Autopsy, ingest and analyze storage evidence data sources, examine deleted files and metadata structures, and generate a comprehensive forensic report.

TOOLS / REQUIREMENTS

- **Forensic Platform:** Autopsy 4.23.1 (64-bit) / The Sleuth Kit (TSK) engine
- **Host Operating System:** Microsoft Windows 11
- **Evidence Source:** Virtual Hard Disk volume (VHD.vhd / New Volume (D:) 2.0 GB)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A forensic investigation case was initiated in Autopsy to analyze a storage volume containing deleted artifacts. The examiner configured formal case information, ingested the data source, ran automated analysis modules, identified deleted credential remnants from unallocated space, inspected low-level Master File Table (MFT) records using Sleuth Kit `istat` integration, and generated an exportable forensic report.

EVIDENCE / INPUT

- **Case Name:** CASE-2026-04(AUTOPSY)
- **Case Number:** 04
- **Examiner Details:** SEKHAR REDDY (Phone: 9392746218, Email: sekharnandireddy338@gmail.com)
- **Ingested Data Source:** VHD.vhd_1 Host / New Volume (D:) 2.0 GB
- **Target Deleted File:** /img_VHD.vhd/vol_vol2/User_credentials.txt.txt (Size: 18 bytes, Unallocated, MFT Entry 45, Sequence 2)
- **Target Report Format:** Excel Spreadsheet (Excel.xlsx)

PROCEDURE

Step 1 — Configuring Case Information and Examiner Details

Autopsy 4.23.1 was launched and a new forensic case was initialized. In the **Optional Information** window, case details were configured: Case Number 04, Examiner Name SEKHAR REDDY, Phone 9392746218, Email sekhnandireddy338@gmail.com.

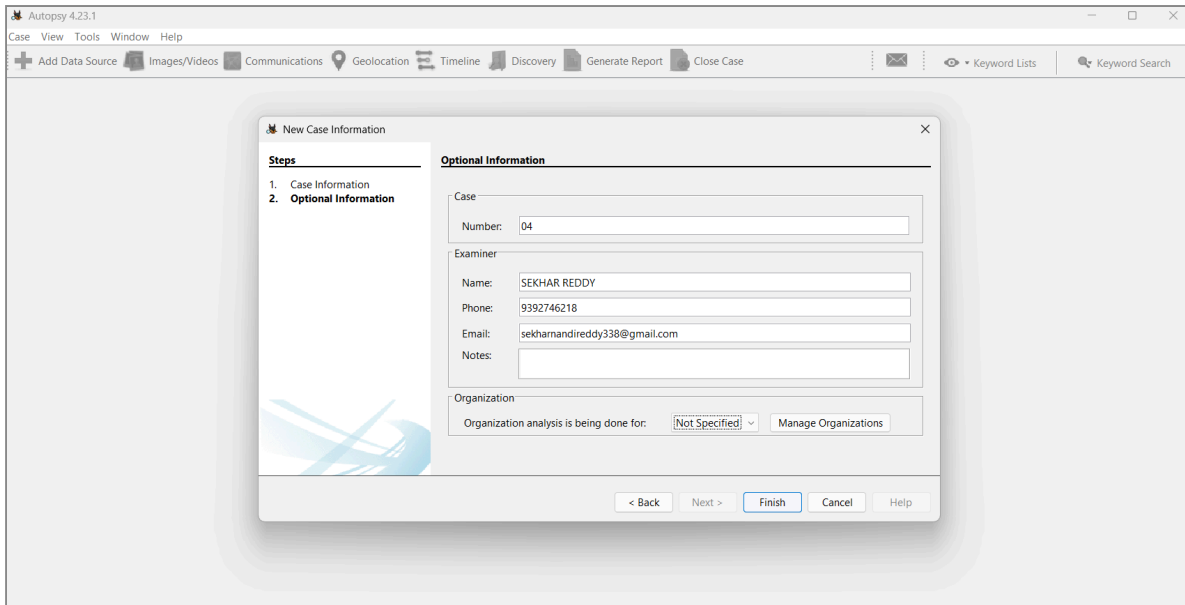


Figure 5.1: Autopsy new case optional information window

Step 2 — Selecting and Adding Storage Data Source

The *Add Data Source* wizard was executed. Under the Local Disk option, the target evidence drive New Volume (D:) 2.0 GB was selected with timezone set to (GMT+5:30) Asia/Calcutta.

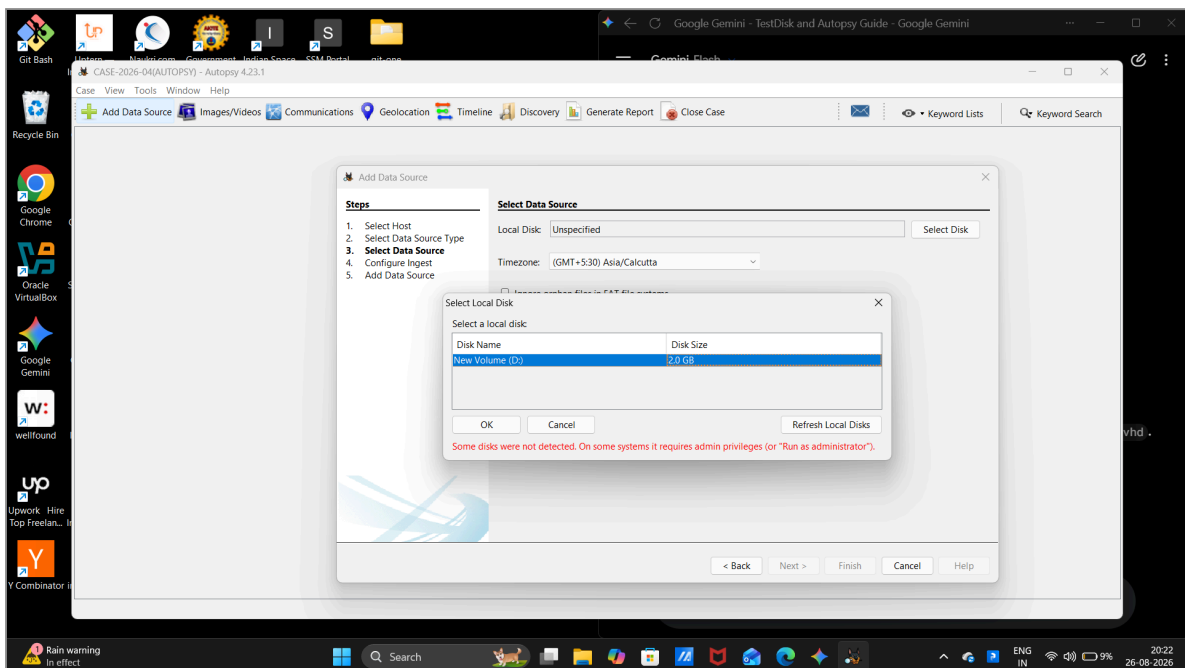


Figure 5.2: Adding local disk data source New Volume (D:)

Step 3 — Confirming Data Source Ingest into Local Database

The data source was committed and ingested into the SQLite case database, initiating automated ingest modules.

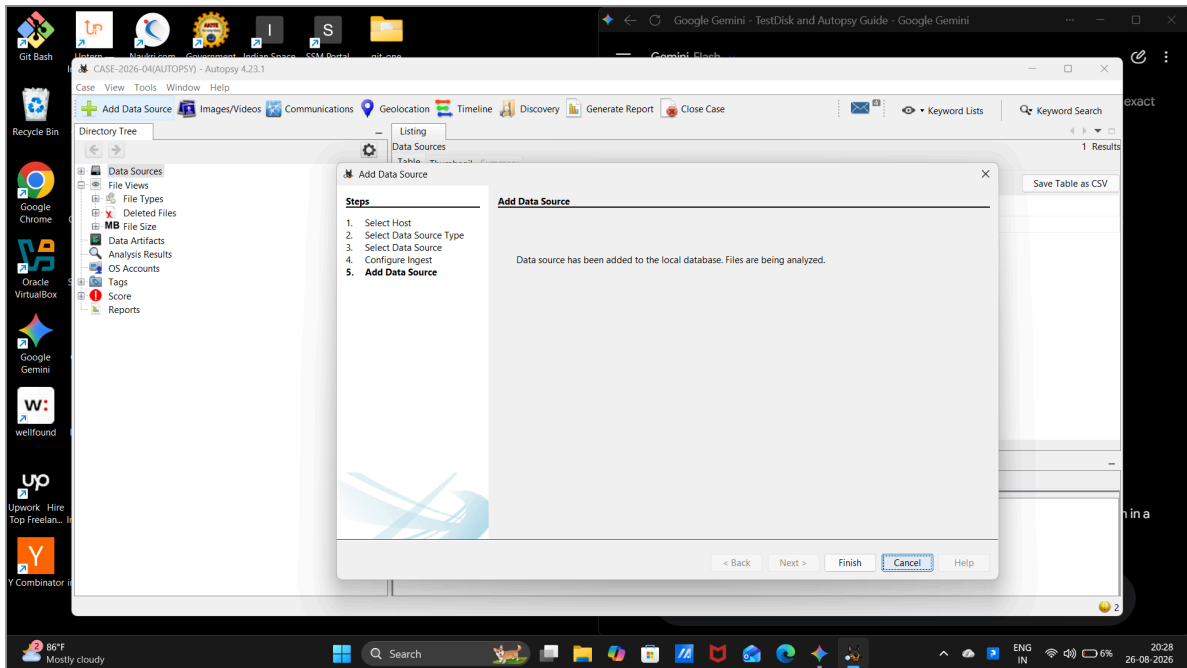


Figure 5.3: Data source ingest confirmation dialog

Step 4 — Exploring Directory Tree and Data Source Hierarchy

The Autopsy Directory Tree was navigated to inspect the ingested data source hierarchy, confirming active indexing of VHD.vhd_1 Host.

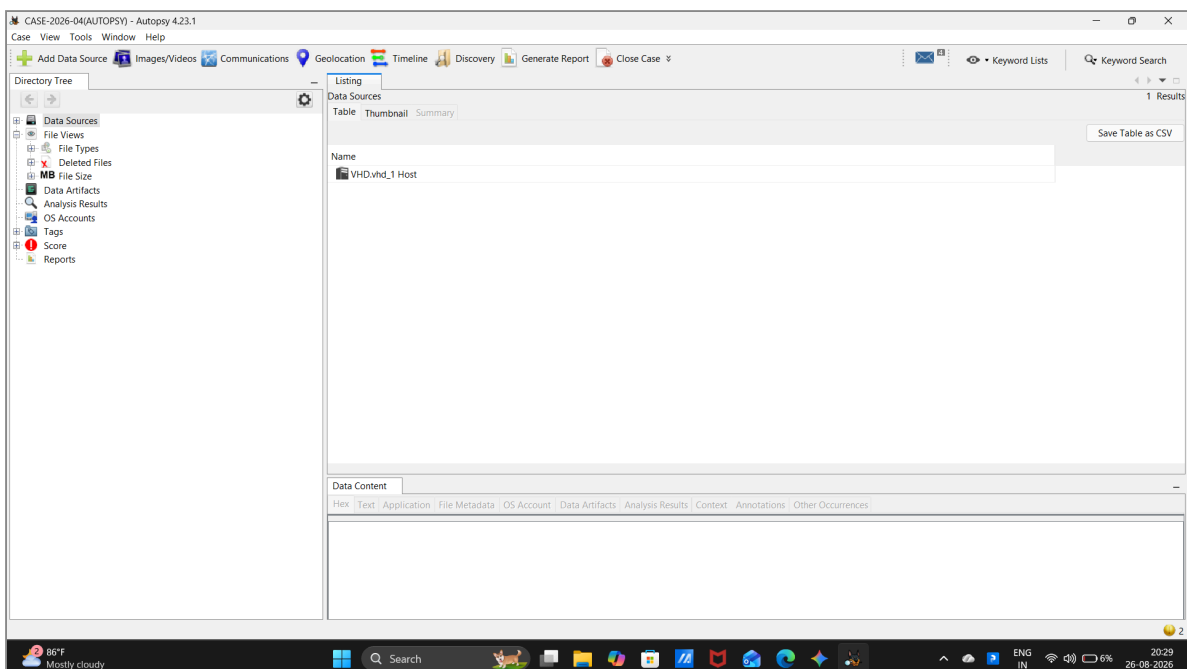


Figure 5.4: Directory tree displaying indexed data sources

Step 5 — Analyzing Deleted Files and Text Artifacts

Under **File Views -> Deleted Files -> All (1)**, the deleted artifact `User_credentials.txt.txt` (18 bytes, Unallocated) was identified. The lower Data Content pane (Text view) revealed the recovered credential payload: `admin:password@123`.

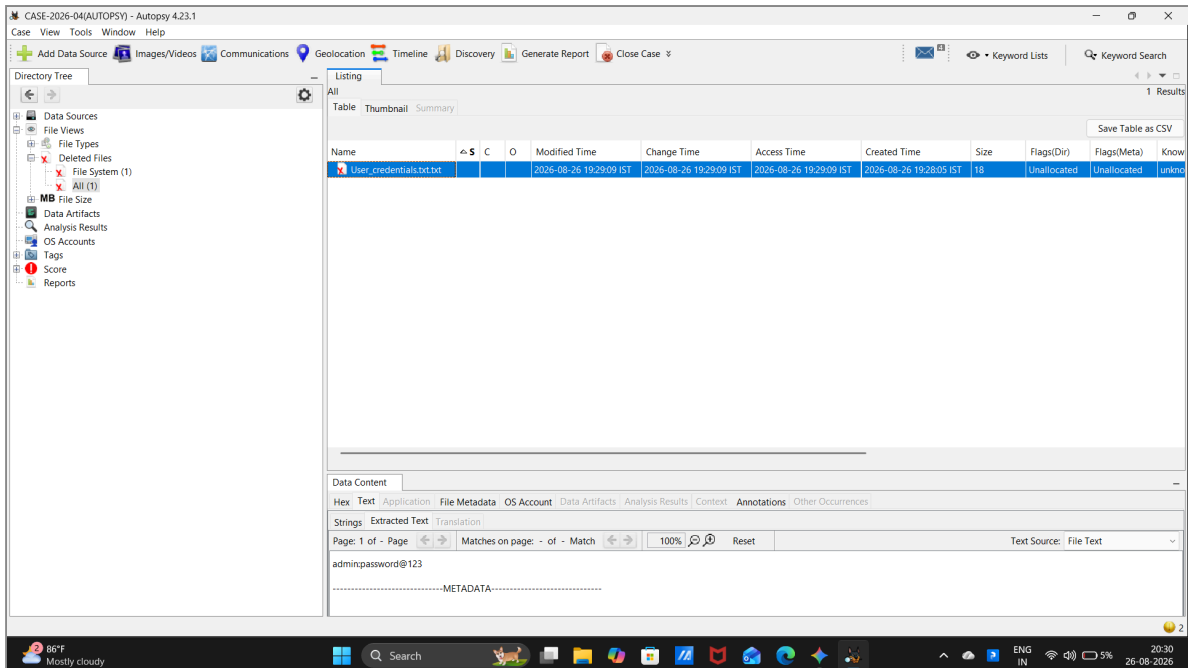


Figure 5.5: Deleted files view showing recovered text payload

Step 6 — Examining File Metadata and Sleuth Kit istat Output

The **File Metadata** tab was selected to analyze low-level filesystem attributes integrated from The Sleuth Kit **istat** tool:

- **File Name Allocation:** Unallocated
- **Metadata Allocation:** Unallocated
- **Modified / Accessed / Changed:** 2026-08-26 19:29:09 IST
- **Created:** 2026-08-26 19:28:05 IST
- **MFT Entry Header:** Entry: 45 Sequence: 2
- **Flags:** Archive

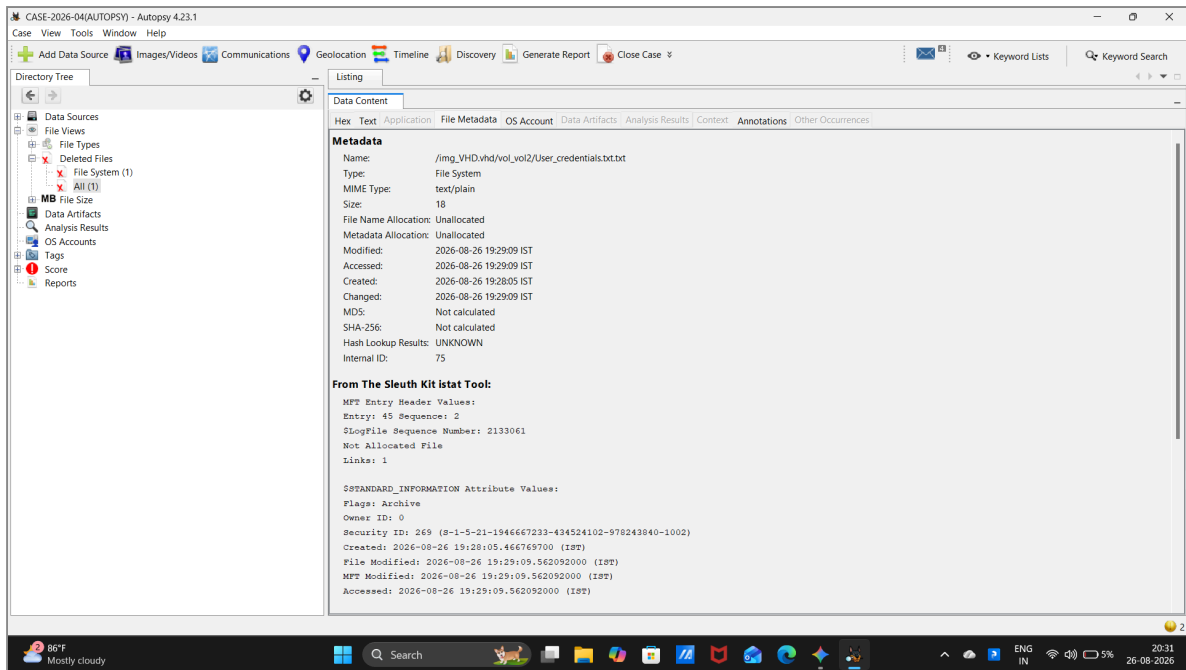


Figure 5.6: File metadata tab showing Sleuth Kit istat attributes

Step 7 — Configuring Forensic Report Generation Module

The *Generate Report* tool was opened from the toolbar. The **Excel Report** module was selected to create a structured spreadsheet report of findings and tagged items.

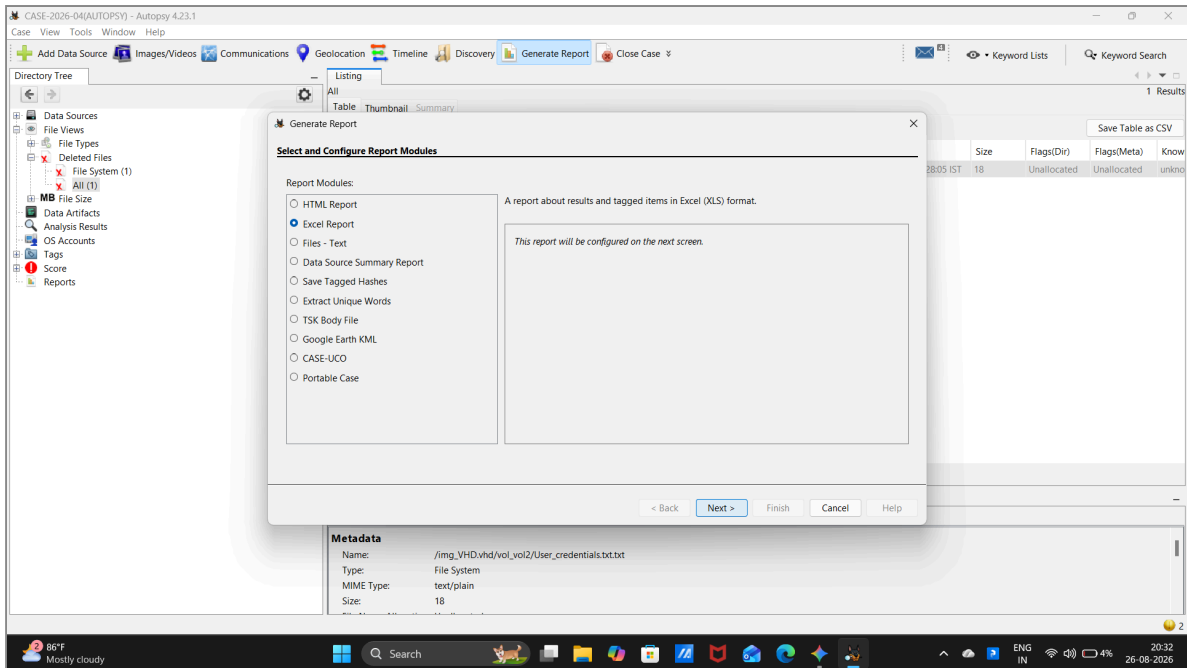


Figure 5.7: Generate Report dialog selecting Excel Report

Step 8 — Selecting Target Data Source for Reporting

In the report configuration wizard, the ingested data source VHD.vhd was selected for inclusion.

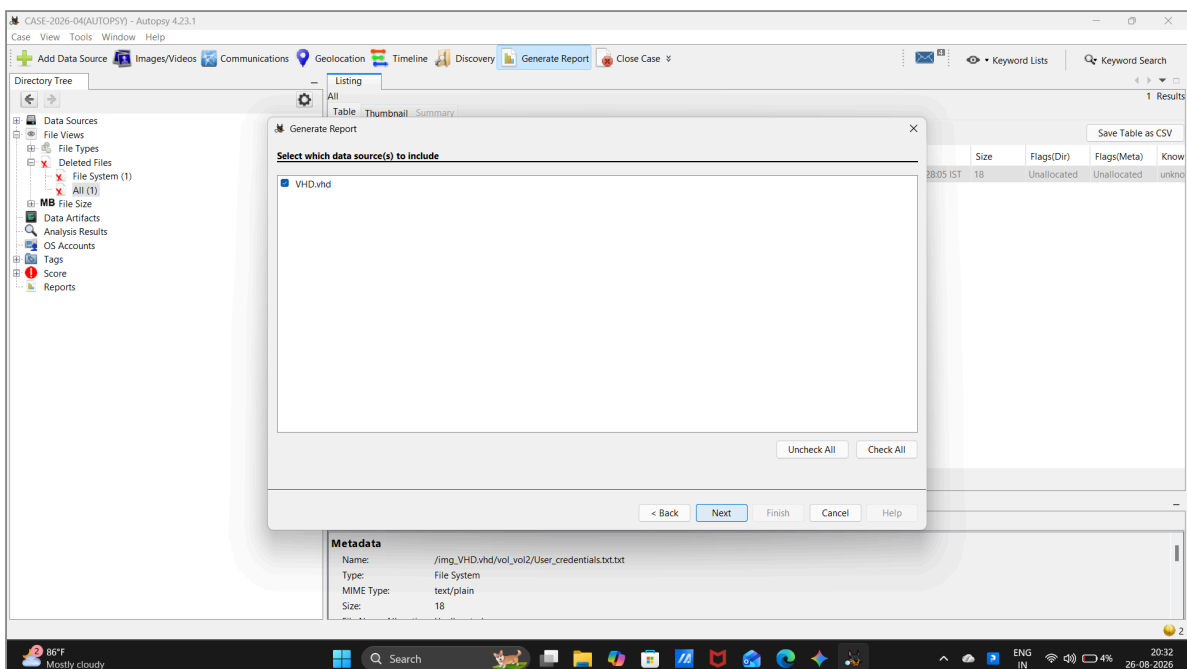


Figure 5.8: Selecting VHD data source for report inclusion

Step 9 — Executing Report Generation Pipeline

The report generation engine processed the case database and compiled artifact tables.

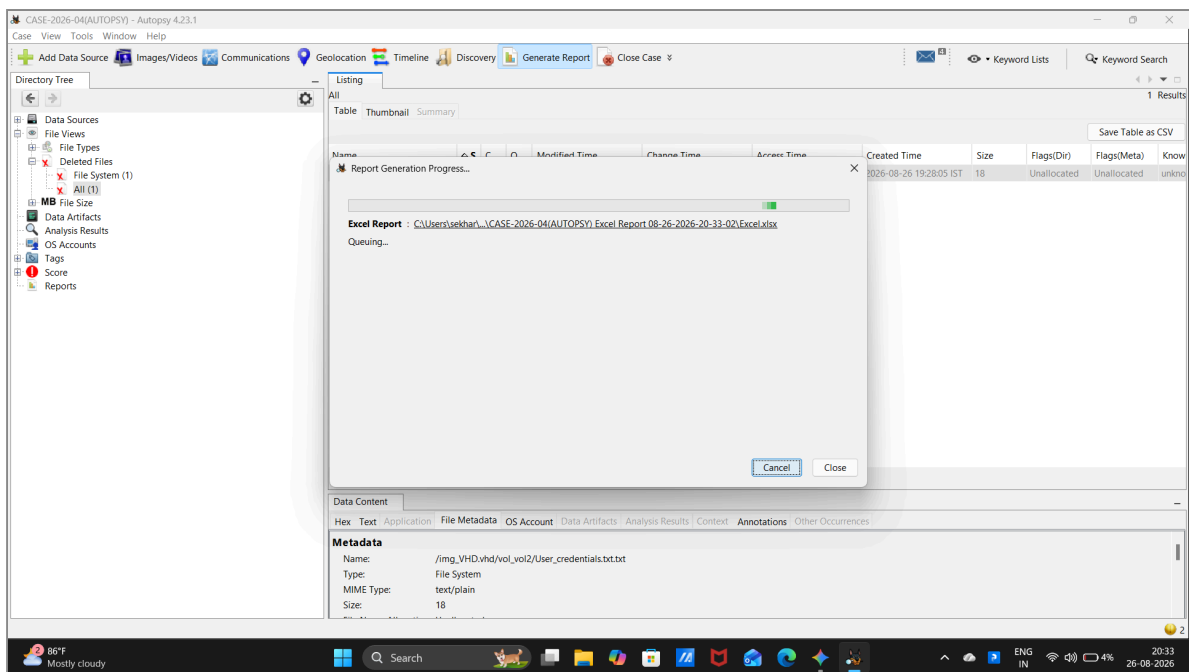


Figure 5.9: Report generation progress dialog

Step 10 — Verifying Final Report Generation Completion

The report generation pipeline concluded with status Complete. The generated report path was confirmed: CASE-2026-04(AUTOPSY) Excel Report 08-26-2026-20-33-02\Excel.xlsx.

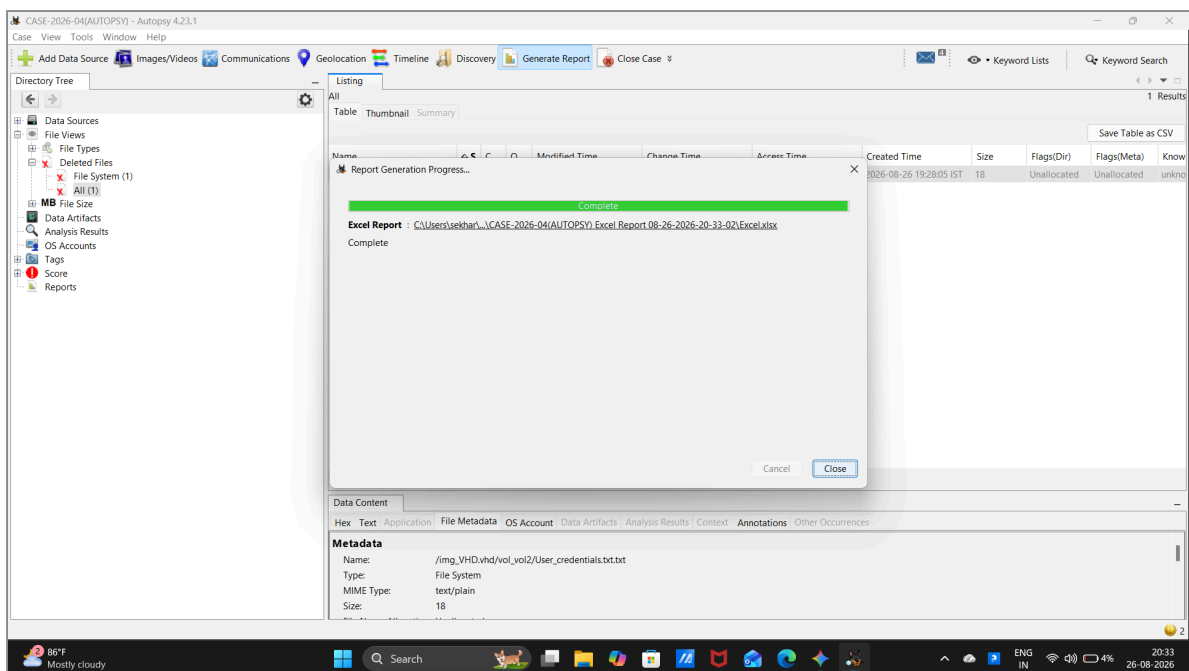


Figure 5.10: Report generation complete confirmation

OBSERVATIONS

1. Autopsy successfully created case CASE-2026-04(AUTOPSY) with examiner metadata.
2. Ingestion of data source VHD.vhd_1 Host indexed volume partitions and file structures.
3. Deleted file view isolated unallocated record User_credentials.txt.txt (18 bytes).

4. Content viewer extracted string `admin:password@123`.
5. Sleuth Kit `istat` metadata verified MFT Entry 45, Sequence 2, and timestamp records.
6. Report generation generated `Excel.xlsx` in the case reports directory.

FINDINGS

- **Artifact Carving and Recovery:** Autopsy parsed unallocated space on the NTFS volume to identify and preview deleted file contents without requiring full sector imaging.
- **Filesystem Metadata Auditing:** Integration with The Sleuth Kit provided low-level MFT inode and timestamp tracking essential for forensic reconstruction.
- **Reporting Soundness:** The automated report generation module compiled findings into a standardized audit format.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The forensic case was created in Autopsy, the disk image was ingested and analyzed, deleted credentials were recovered, and a comprehensive Excel report was generated.

CONCLUSION

Autopsy effectively managed the forensic case lifecycle from data source ingestion and unallocated space analysis to MFT metadata inspection and automated report compilation.

REFERENCES

- Autopsy User Documentation (Basis Technology / Sleuth Kit)
- Digital Forensics Laboratory Manual: Use Autopsy to create a case and import evidence